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DYADIC DENOMINATOR BOUNDS AND A TRANSFER PRINCIPLE FOR THE PHASE CONSTANTS OF JACOBI ZEROS

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ABSTRACT. A formal large-degree phase expansion associated with the zeros of the Jacobi polynomial $P_n^{(\alpha,\beta)}$ contains a sequence of constants $\kappa_1, \kappa_2, \dots$, each of which is a polynomial in α^2 and β^2 with rational coefficients; for instance $\kappa_1 = (\alpha^2 - \beta^2)/4$. This paper asks a simple arithmetic question: which primes, and to which powers, occur in the denominators of these coefficients? We prove that the odd part of the common denominator of κ_r divides $\operatorname{lcm}(1, 3, \dots, 2r-1)$, and that the power of 2 is at most 2^{E_r} with $E_r = 3r - 1 + \nu_2((r-1)!)$, where ν_2 is the 2-adic valuation. The study of the leading coefficient leads to the formula

$$\nu_2\left(\sum_{j=0}^{m} \binom{m}{j} \frac{1}{2j+1}\right) = m + \nu_2(m+1) \qquad (m \geq 0),$$

which we prove by showing that the series $\sum_{k\geq 0} k!/(2k+1)!!$, whose real sum is $\pi/2$, converges to 0 in the field $\mathbb{Q}_2$ of 2-adic numbers. Finally, we show that the statement "the exponent E_r is attained for every r" is equivalent to an integrality property of the Taylor coefficients of a single explicit power series in one variable. That property is established in a separate paper, and is confirmed here independently, by exact computation, for $r \leq 48$.

1. INTRODUCTION

Let $P_n^{(\alpha,\beta)}$ be the Jacobi polynomial of degree n with parameters $\alpha, \beta > -1$, in the standard normalisation of [1, 2]. Its n zeros lie in $(-1, 1)$; we write them as $x_{n,k} = \cos\theta_{n,k}$ with $0 < \theta_{n,1} < \cdots < \theta_{n,n} < \pi$. Throughout the paper

$$\mathcal{N} = n + \frac{\alpha+\beta+1}{2}, \qquad A = \alpha^2, \qquad B = \beta^2.$$

The classical first-order description of the zeros (cf. [1, Ch. 8]) is $\mathcal{N}\theta_{n,k} \approx \left(k + \frac{\alpha}{2} - \frac{1}{4}\right)\pi$. It is the first term of a complete asymptotic expansion: transforming the Jacobi differential equation to Liouville normal form and solving the associated phase equation recursively in powers of $\mathcal{N}^{-2}$, one obtains a *phase function*

$$\Xi(\theta) = \mathcal{N}\theta - \left(\frac{\alpha}{2} + \frac{1}{4}\right)\pi + \sum_{r\geq 1} \frac{\Psi_r(\theta)}{\mathcal{N}^{2r-1}}, \tag{1}$$

such that the zeros are, to all orders in $1/\mathcal{N}$, the solutions of $\Xi(\theta_{n,k}) = (k - \frac{1}{2})\pi$. Each Ψ_r is obtained by an integration in θ, and is therefore determined only up to an additive constant. With the natural choice of primitive (the one that is an odd Laurent polynomial in $\cot(\theta/2)$), Ψ_r is a Laurent polynomial in $\cot(\theta/2)$ whose coefficients are polynomials in

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A and B with rational coefficients. The additive constant is then recorded by the value at the midpoint $\theta = \pi/2$ of the interval,

$$\kappa_r = \Psi_r(\pi/2) \in \mathbb{Q}[A, B].$$

We call $\kappa_1, \kappa_2, \dots$ the *phase constants*. The first three are

$$\kappa_1 = \frac{A-B}{4}, \qquad \kappa_2 = \frac{-7(A-B) + 2(A^2-B^2)}{96},$$
$$\kappa_3 = \frac{781(A-B)}{7680} - \frac{19(A^2-B^2)}{384} + \frac{7(A^3-B^3)}{1920} + \frac{AB(A-B)}{128}.$$

Each κ_r changes sign when A and B are exchanged (Remark 2.2), so it is divisible by $A-B$ in $\mathbb{Q}[A,B]$. In particular the phase constants vanish when $\alpha = \pm\beta$, which is why they do not appear in the classical theory of Legendre and Gegenbauer polynomials.

Let $\operatorname{den}\kappa_r$ denote the least positive integer d such that $d\,\kappa_r$ has integer coefficients. The first values are shown in Table 1.

r	$\operatorname{den}\kappa_r$	factorisation	$E_r = 3r - 1 + \nu_2((r-1)!)$	$\operatorname{lcm}(1, 3, \dots, 2r-1)$
1	4	2^2	2	1
2	96	$2^5 \cdot 3$	5	3
3	7680	$2^9 \cdot 15$	9	15
4	86016	$2^{12} \cdot 21$	12	105

TABLE 1. Denominators of the first four phase constants. The power of 2 is exactly 2^{E_r}, and the odd part divides $\operatorname{lcm}(1, 3, \dots, 2r-1)$.

The question studied in this paper is: *what are the prime factors of* $\operatorname{den}\kappa_r$*, and in particular what is the exact power of* 2*?* The odd part turns out to be easy. The power of 2 is the substantial part: its exponent $2, 5, 9, 12, \dots$ is given by $E_r = 3r - 1 + \nu_2((r-1)!)$. The purpose of the present paper is to prove that E_r is an upper bound for every r, to isolate the exact obstruction to its sharpness, and to verify sharpness in a substantial finite range; the proof of sharpness for every r is completed in [17].

We first collect everything that is used from elementary 2-adic arithmetic; a standard reference is [3].

- For a nonzero rational number $q = 2^e\, a/b$ with a, b odd integers, $\nu_2(q) = e$ is the 2-*adic valuation* of q; $\nu_2(0) = +\infty$. It satisfies $\nu_2(pq) = \nu_2(p) + \nu_2(q)$ and $\nu_2(p+q) \geq \min\big(\nu_2(p), \nu_2(q)\big)$, *with equality when the two valuations are different.* This last rule is used repeatedly.
- $\mathbb{Q}_2$ is the field of 2-adic numbers, the completion of $\mathbb{Q}$ for the absolute value $|q|_2 = 2^{-\nu_2(q)}$, and $\mathbb{Z}_2 = \{q \in \mathbb{Q}_2 : \nu_2(q) \geq 0\}$ is the ring of 2-adic integers. A rational number belongs to $\mathbb{Z}_2$ exactly when its denominator is odd. The *units* of $\mathbb{Z}_2$, written $\mathbb{Z}_2^\times$, are the elements of valuation 0. A series $\sum_k c_k$ converges in $\mathbb{Q}_2$ if and only if $\nu_2(c_k) \to +\infty$.
- $s_2(m)$ is the number of digits 1 in the binary expansion of the non-negative integer m. It is subadditive: $s_2(a+b) \leq s_2(a) + s_2(b)$.
- *Legendre's formula*: $\nu_2(m!) = m - s_2(m)$.
- *Kummer's theorem* [4, 5]: $\nu_2\binom{2m}{m} = s_2(m)$.

- $[t^k]f(t)$ is the coefficient of t^k in a power series or polynomial f; similarly $[A^iB^j]$ for polynomials in two variables.
- $(a)_k = a(a+1)\cdots(a+k-1)$ is the Pochhammer symbol, and $(2k+1)!! = 1\cdot 3\cdot 5\cdots(2k+1)$.
- $\mathcal{B}_{2k}$ are the Bernoulli numbers ($\mathcal{B}_2 = \frac{1}{6}$, $\mathcal{B}_4 = -\frac{1}{30}, \dots$). By the von Staudt–Clausen theorem [6, 7, 8], $\nu_2(\mathcal{B}_{2k}) = -1$ for $k \geq 1$.

Three quantities built from these appear throughout:

(2)
$$H_r = r+\nu_2\big((r-1)!\big), \qquad E_r = (2r-1)+H_r = 3r-1+\nu_2\big((r-1)!\big), \qquad \omega(s) = \left\lceil \frac{s}{2} \right\rceil + \nu_2(s).$$

It is convenient to use, next to A and B, the shifted variables

$$\mu_\alpha = A - \tfrac{1}{4} = \alpha^2 - \tfrac{1}{4}, \qquad \mu_\beta = B - \tfrac{1}{4} = \beta^2 - \tfrac{1}{4},$$

which are the combinations in which the parameters enter the differential equation. A polynomial in (A, B) is also a polynomial in (μ_α, μ_β), and conversely.

Next we present the main results of the paper, which shall be proved in the following sections. Precisely, κ_r is defined in this paper by formula (7) of Section 2; its relation with (1) is discussed at the end of this Introduction. The first two theorems bound the denominator.

Theorem 1.1 (odd part of the denominator)**.** *For every $r \geq 1$, the odd part of* $\operatorname{den} \kappa_r$ *divides* $\operatorname{lcm}(1, 3, \dots, 2r-1)$.

Theorem 1.2 (power of 2: upper bound)**.** *For every $r \geq 1$, all coefficients of the polynomial $2^{E_r}\kappa_r$, written in the variables μ_α, μ_β, are 2-adic integers: $2^{E_r}\kappa_r \in \mathbb{Z}_2[\mu_\alpha, \mu_\beta]$.*

The next two theorems are arithmetic statements of independent interest. They arise from the coefficient of the highest power μ_α^r in κ_r, which is a multiple of the sum in Theorem 1.4.

Theorem 1.3 (a 2-adic vanishing)**.** *The series*

$$\sum_{k\geq 0} \frac{k!}{(2k+1)!!} = \sum_{k\geq 0} \frac{2^k}{(2k+1)\binom{2k}{k}}$$

converges in $\mathbb{Q}_2$, and its sum there is 0.

In the real numbers the same series converges to $\pi/2$. Indeed, $k!/(2k+1)!! = 2^{-k}\int_0^1 (1-u^2)^k\,du$, and summing the geometric series under the integral sign gives $\int_0^1 2/(1+u^2)\,du = \pi/2$.

Theorem 1.4 (valuation of a binomial sum)**.** *For every integer $m \geq 0$,*

$$\nu_2\Big(\sum_{j=0}^{m} \binom{m}{j} \frac{1}{2j+1}\Big) = m + \nu_2(m+1).$$

For example, $m = 3$ gives $1 + 1 + \frac{3}{5} + \frac{1}{7} = \frac{96}{35}$ and $\nu_2(96) = 5 = 3 + \nu_2(4)$.

The last result concerns the question whether the exponent E_r of Theorem 1.2 is actually attained, as it is in Table 1. The relevant coefficient is that of the monomial A in κ_r,

(3)
$$\ell_r = [A^1B^0]\,\kappa_r = \frac{\partial \kappa_r}{\partial A}\Big|_{A=B=0}, \qquad \ell_1 = \frac{1}{4}, \quad \ell_2 = -\frac{7}{96}, \quad \ell_3 = \frac{781}{7680}, \quad \ell_4 = -\frac{28421}{86016}.$$

Since $A = B = 0$ is the Legendre case $\alpha = \beta = 0$, the number ℓ_r measures the first-order response of κ_r when one moves away from the Legendre polynomials in the direction of A. We call

$$\nu_2(\ell_r) = -E_r \qquad \text{for every } r \geq 1 \tag{4}$$

the *sharp law*. The difficulty in proving (4) is that ℓ_r arises as a sum of very many terms, and the valuation of the sum is strictly larger than the smallest valuation among its terms: 2-adic cancellation takes place. Our result is that (4) can be restated, without any loss, as a property of one power series.

Theorem 1.5 (transfer to one power series)**.** *Let $g_n \in \mathbb{Q}(i)$, $n \geq 1$, be the numbers defined in (19) below, so that $\ell_r = 2^{-(2r-1)} \operatorname{Im} g_{2r-1}$, and define the even power series*

$$\sum_{m\geq 0} w_m t^{2m} = \text{even part of } \frac{\sinh 2t}{t^2} \sum_{n\geq 1} \frac{\operatorname{Im} g_n}{(n-1)!}\, t^n = 1 + \frac{t^2}{12} + \frac{t^4}{64} - \frac{143\, t^6}{80640} + \cdots . \tag{5}$$

Then, for every $r \geq 1$,

$$\nu_2(\ell_j) \geq -E_j \ \text{ for all } j \leq r \quad \Longleftrightarrow \quad \nu_2(w_m) \geq -2\nu_2\big((2m)!\big) \ \text{ for all } m \leq r-1,$$

and, when these inequalities hold, $\nu_2(\ell_r) = -E_r$ if and only if $\nu_2(w_{r-1}) = -2\nu_2((2r-2)!)$. Consequently the sharp law (4) is equivalent to

$$\big((2m)!\big)^2\, w_m \in \mathbb{Z}_2^{\times} \qquad \text{for every } m \geq 0. \tag{W}$$

Property (W) is proved in the companion paper [17], by an analytic computation of the series (5) in closed form; together with Theorem 1.5 this establishes the sharp law (4) for every r. No result of the present paper depends on [17]. Independently of it, we verify (W) here by exact computation for $m \leq 46$, that is, by Theorem 1.5, the sharp law for $r \leq 47$; a direct computation of ℓ_{48} gives the case $r = 48$ as well (Section 5).

We close this Introduction by making precise the status of the statements and the organisation of the paper. Two statements in this paper rest on exact computation in a finite range rather than on a proof valid for all r; we state them here so that there is no ambiguity.

(a) *The definition of κ_r.* Our proofs use the definition (7) of κ_r in terms of the logarithm of an explicit double hypergeometric series. That this coincides with the constant $\Psi_r(\pi/2)$ of the phase expansion (1) has been checked in exact rational arithmetic for $r \leq 8$, against an independent computation based on the phase recursion. All theorems are statements about the numbers defined by (7). Accordingly, their interpretation as statements about the constants of the phase expansion (1) for $r > 8$ is conditional on the identification between (7) and (1) for those r.

(b) *The leading coefficient of κ_r.* The identification of the coefficient of μ_α^r in κ_r with a multiple of the binomial sum of Theorem 1.4 (formula (14)) has been checked exactly for $r \leq 20$. Theorems 1.3 and 1.4 themselves are proved for all m and do not depend on it.

Everything labelled Theorem, Lemma, Proposition or Corollary is proved in full, with any dependence on (b) stated explicitly; statements obtained by computation are labelled "Computational observation". All computations are exact (rational or Gaussian-rational arithmetic); no floating-point arithmetic has been used.

Section 2 defines κ_r, proves a division-free recurrence with integer coefficients, and deduces Theorems 1.1 and 1.2. Section 3 proves Theorems 1.3 and 1.4 and applies them

to the leading coefficient; it can be read independently of the rest. Section 4 proves Theorem 1.5. Section 5 reports the computations. Appendix A records an integral representation related to [17].

2. The phase constants and an integral recurrence

We begin with the precise definition of the phase constants. For $0 < \theta < \pi$ put

$$x = -\frac{1}{e^{i\theta} - 1}, \qquad y = \frac{1}{e^{i\theta} + 1}, \qquad Z = 2\mathcal{N},$$

and let

$$a_k(\nu) = \frac{(\frac{1}{2} + \nu)_k (\frac{1}{2} - \nu)_k}{(-2)^k k!} = \frac{1}{8^k k!} \prod_{j=0}^{k-1} \bigl(4\nu^2 - (2j+1)^2\bigr)$$

be, up to normalisation, the coefficients of Hankel's expansion of the Bessel functions [2, Sec. 10.17]. The large-degree asymptotics of $P_n^{(\alpha,\beta)}(\cos\theta)$ is governed by the formal inverse-factorial series

$$F = \sum_{n \geq 0} \frac{C_n}{(Z-1)(Z-2)\cdots(Z-n)}, \qquad C_n = \sum_{l=0}^{n} a_l(\alpha)\, a_{n-l}(\beta)\, (2x)^l (2y)^{n-l}, \tag{6}$$

which is our starting point. Each term of (6) can be expanded in powers of Z^{-1}, so the logarithm of F is a formal power series

$$L = \log F = \sum_{n \geq 1} L_n\, Z^{-n},$$

whose coefficients L_n are polynomials in x and y without constant term, with coefficients in $\mathbb{Q}[A, B]$ (Theorem 2.3 below). The corrections to the phase in (1) are carried by the imaginary part of L. At the midpoint $\theta = \pi/2$ we have $e^{i\theta} = i$ and

$$x_0 = \frac{1+i}{2}, \qquad y_0 = \frac{1-i}{2}.$$

Definition 2.1. For $r \geq 1$,

$$\kappa_r = 2^{-(2r-1)} \operatorname{Im} L_{2r-1}(x_0, y_0) \;\in\; \mathbb{Q}[A, B], \tag{7}$$

where the imaginary part is taken treating A and B as real variables.

Remark 2.2 (antisymmetry). The series (6) is unchanged when (α, x) and (β, y) are interchanged, so $L_n(x, y; A, B) = L_n(y, x; B, A)$. The coefficients of L_n are rational and $y_0 = \overline{x_0}$, hence $L_n(x_0, y_0; A, B) = \overline{L_n(x_0, y_0; B, A)}$ for real A, B. Taking imaginary parts, $\kappa_r(B, A) = -\kappa_r(A, B)$.

The factor $2^{-(2r-1)}$ converts powers of $Z^{-1} = (2\mathcal{N})^{-1}$ into powers of $\mathcal{N}^{-1}$. As explained in item (a) of the Introduction, the agreement of (7) with the constants of (1) has been verified for $r \leq 8$; the values of $\kappa_1, \kappa_2, \kappa_3$ displayed in the introduction are obtained from both.

The key to the arithmetic of κ_r is that the logarithmic derivatives of F satisfy a recurrence in which no division occurs. Write

$$\Theta_x = x\frac{\partial}{\partial x}, \qquad \Theta_y = y\frac{\partial}{\partial y}, \qquad U = \Theta_x L, \qquad V = \Theta_y L, \qquad T = U + V,$$

and expand $U=\sum_{n\ge1}U_nZ^{-n}$, $V=\sum_{n\ge1}V_nZ^{-n}$, $T_n=U_n+V_n$. Note that Θ_x and Θ_y multiply the monomial x^cy^d by c and by d respectively.

Theorem 2.3 (integral recurrence)**.** *We have $U_1=x\mu_\alpha$, $V_1=y\mu_\beta$, and for $n\ge1$*

$$U_{n+1}=\Theta_xT_n+\sum_{\substack{a+b=n\\a,b\ge1}}U_aT_b-x\Big(\Theta_xU_n+U_n+\sum_{\substack{a+b=n\\a,b\ge1}}U_aU_b\Big), \tag{8}$$

and the same formula for V_{n+1} with the roles of (x,U,μ_α) and (y,V,μ_β) exchanged. Consequently, for every $n\ge1$:

(i) *U_n and V_n are polynomials in x,y,μ_α,μ_β with* integer *coefficients;*
(ii) *every monomial $x^cy^d\mu_\alpha^a\mu_\beta^b$ occurring in U_n or V_n satisfies $1\le a+b\le c+d\le n$.*

Proof. Step 1: F is a double hypergeometric series. Since $1/\big((Z-1)\cdots(Z-n)\big)=(-1)^n/(1-Z)_n$ and $(-1)^l(2x)^la_l(\alpha)=(\frac12-\alpha)_l(\frac12+\alpha)_lx^l/l!$, the series (6) is

$$F=\sum_{c,d\ge0}F_{c,d}\,x^cy^d,\qquad F_{c,d}=\frac{(\frac12-\alpha)_c(\frac12+\alpha)_c(\frac12-\beta)_d(\frac12+\beta)_d}{(1-Z)_{c+d}\;c!\;d!}, \tag{9}$$

an identity of formal power series in x,y whose coefficients are rational functions of Z.

Step 2: a partial differential equation. From (9),

$$\frac{F_{c+1,d}}{F_{c,d}}=\frac{(\frac12-\alpha+c)(\frac12+\alpha+c)}{(1-Z+c+d)(c+1)}=\frac{(c+\frac12)^2-A}{(c+1+d-Z)(c+1)}.$$

Multiplying out and comparing the coefficients of $x^{c+1}y^d$, this says exactly that

$$\Theta_x\big(\Theta_x+\Theta_y-Z\big)F=x\Big(\big(\Theta_x+\tfrac12\big)^2-A\Big)F, \tag{10}$$

and symmetrically in y with B.

Step 3: the equation for the logarithm. Substitute $F=e^L$. Then $\Theta_xF=UF$, $(\Theta_x+\Theta_y)F=TF$, $\Theta_x(TF)=(\Theta_xT+TU)F$ and $(\Theta_x+\frac12)^2F=(\Theta_xU+U^2+U+\frac14)F$. Dividing (10) by F and using $A-\frac14=\mu_\alpha$,

$$ZU=\Theta_xT+UT-x\big(\Theta_xU+U^2+U-\mu_\alpha\big). \tag{11}$$

The coefficient of Z^{-n} on the left is U_{n+1}. Comparing the coefficients of Z^0 gives $U_1=x\mu_\alpha$, and comparing those of Z^{-n}, $n\ge1$, gives (8).

Step 4: integrality and support. Both hold for $U_1=x\mu_\alpha$ and $V_1=y\mu_\beta$. The right-hand side of (8) involves only Θ_x, multiplication by x, sums and products; all of these preserve integer coefficients, which proves (i) by induction. For (ii), call $c+d$ the degree and $a+b$ the μ-degree of a monomial. In a product both add, so the inequalities $a+b\le c+d$ and $c+d\le n$ (with n additive along U_aT_b, $a+b=n$) are preserved; Θ_x changes neither; the multiplication by x raises the degree by one, and it is applied to terms of degree at most n. Hence every monomial of U_{n+1} has $1\le a+b\le c+d\le n+1$. □

Since $T=(\Theta_x+\Theta_y)L$ and L_n has no constant term, L_n is recovered from T_n by dividing the coefficient of each monomial x^cy^d by $c+d$:

$$[x^cy^d]\,L_n=\frac{[x^cy^d]\,T_n}{c+d}. \tag{12}$$

Example 2.4 (the case $r = 1$)**.** Here $T_1 = x\mu_\alpha + y\mu_\beta$, all monomials have $c + d = 1$, so $L_1 = x\mu_\alpha + y\mu_\beta$ and $L_1(x_0, y_0) = \frac{1+i}{2}\mu_\alpha + \frac{1-i}{2}\mu_\beta$. Its imaginary part is $\frac{1}{2}(\mu_\alpha - \mu_\beta) = \frac{1}{2}(A - B)$, and (7) gives $\kappa_1 = \frac{1}{4}(A - B)$.

Where the denominators come from. By Theorem 2.3 and (12), κ_r is an integer combination of monomials $\mu_\alpha^a \mu_\beta^b$ multiplied by the numbers

$$2^{-(2r-1)} \operatorname{Im} \frac{x_0^c\, y_0^d}{c+d}, \qquad 1 \le c + d \le 2r - 1. \tag{13}$$

Thus there are exactly three sources of denominators: the factor $2^{-(2r-1)}$; the division by $s = c + d$; and the powers of $x_0 = \frac{1+i}{2}$, $y_0 = \frac{1-i}{2}$. The next lemma computes the last two exactly.

Lemma 2.5. *Let $c, d \ge 0$ with $s = c + d \ge 1$. If $\operatorname{Im}(x_0^c y_0^d) \ne 0$, then*

$$\nu_2\Big(\operatorname{Im} \frac{x_0^c y_0^d}{s}\Big) = -\omega(s), \qquad \omega(s) = \Big\lceil \frac{s}{2} \Big\rceil + \nu_2(s).$$

Proof. Write $k = \min(c, d)$ and $m = |c - d|$, so that $s = 2k + m$. Since $(1 + i)(1 - i) = 2$,

$$x_0^c y_0^d = 2^{-s}(1 + i)^c (1 - i)^d = 2^{-s+k}\, \eta^m, \qquad \eta = 1 + i \text{ or } 1 - i.$$

Now $\eta^m = 2^{m/2} e^{\pm im\pi/4}$, so $\operatorname{Im} \eta^m = \pm 2^{m/2} \sin(m\pi/4)$, which is 0 if $4 \mid m$, equals $\pm 2^{(m-1)/2}$ if m is odd, and equals $\pm 2^{m/2}$ if $m \equiv 2 \pmod 4$. If m is odd, then s is odd and

$$\nu_2\big(\operatorname{Im} x_0^c y_0^d\big) = -s + k + \frac{m-1}{2} = -\frac{s+1}{2} = -\Big\lceil \frac{s}{2} \Big\rceil.$$

If $m \equiv 2 \pmod 4$, then s is even and $\nu_2(\operatorname{Im} x_0^c y_0^d) = -s + k + \frac{m}{2} = -\frac{s}{2} = -\lceil s/2 \rceil$. Dividing by s subtracts $\nu_2(s)$ in both cases. □

Lemma 2.6. *For every $r \ge 1$, $\max\limits_{1 \le s \le 2r-1} \omega(s) \le H_r = r + \nu_2\big((r-1)!\big)$.*

Proof. By Legendre's formula, $H_r = r + \big(r - 1 - s_2(r-1)\big) = 2r - 1 - s_2(r-1)$. If s is odd, then $\nu_2(s) = 0$ and $\omega(s) = \frac{s+1}{2} \le r \le H_r$. If s is even, then $s \le 2r - 2$ and $\nu_2(s) \le \log_2 s$, so $\omega(s) \le (r-1) + \log_2(2r-2)$; with $m = r - 1$ it suffices to show that

$$\log_2 m + s_2(m) \le m.$$

Since $s_2(m) \le \log_2 m + 1$, this holds as soon as $2 \log_2 m + 1 \le m$, that is for $m \ge 7$. For the remaining cases $r = 1, \dots, 7$ one checks directly that $H_r = 1, 2, 4, 5, 8, 9, 11$, while $\max_{s \le 2r-1} \omega(s) = 1, 2, 4, 4, 7, 7, 8$. □

With these two lemmas we can now prove Theorems 1.1 and 1.2.

Proof of Theorem 1.2. By (13), every coefficient of κ_r in the variables μ_α, μ_β is an integer combination of numbers of valuation at least $-(2r-1) - \omega(s)$ with $s \le 2r - 1$ (Lemma 2.5), hence at least $-(2r-1) - H_r = -E_r$ (Lemma 2.6). □

Proof of Theorem 1.1. In (13) the numbers $x_0^c y_0^d$ and $2^{-(2r-1)}$ have denominators that are powers of 2, so odd primes can enter only through the division by $s = c + d \le 2r - 1$. The odd part of any such s is an odd integer not exceeding $2r - 1$, hence divides $\operatorname{lcm}(1, 3, \dots, 2r - 1)$. This bounds the odd part of the denominator of κ_r in the variables (μ_α, μ_β); passing to (A, B) through $\mu_\alpha = A - \frac{1}{4}$, $\mu_\beta = B - \frac{1}{4}$ introduces only powers of 2. □

Remark 2.7. Theorem 1.2 is stated in the variables (μ_α, μ_β), in which the proof is natural. In these variables the bound 2^{E_r} need not be attained: for $2 \le r \le 8$ the exact exponent of 2 in the denominator is $2r + \nu_2((r-1)!)$, which is smaller than E_r by $r-1$ (see Remark 3.8). It is in the original variables (A, B) that the exponent E_r appears exactly, as in Table 1; this is the sharp law (4), treated in Section 4.

3. The leading coefficient and a 2-adic vanishing

This section proves Theorems 1.3 and 1.4. Apart from the motivation, given first, and the application to the phase constants, given at the end, it uses nothing from Section 2.

We begin with the motivation, which is the coefficient of the highest power μ_α^r in κ_r. In the computed range ($r \le 8$), κ_r has degree exactly r in (μ_α, μ_β), and the coefficients of the pure powers μ_α^r and μ_β^r have a closed form. Put

$$I_r = \int_0^1 (1+u^2)^{r-1} du = \sum_{j=0}^{r-1} \binom{r-1}{j} \frac{1}{2j+1}, \qquad \gamma_r = \frac{2\binom{2r}{r}}{(2r-1)\,16^r}.$$

Computational observation 3.1. For $r \le 20$,

$$[\mu_\alpha^r]\,\kappa_r = \gamma_r I_r, \qquad [\mu_\beta^r]\,\kappa_r = -\gamma_r I_r. \tag{14}$$

This was established by exact computation of κ_r from (7): for $r \le 8$ as a polynomial in both variables, and for $r \le 20$ on the line $\beta = \frac{1}{2}$ (where $\mu_\beta = 0$) by interpolation in μ_α. We expect (14) to hold for every r, but we do not prove it. It is the only place where the binomial sum I_r enters the Jacobi problem; the task of this section is to determine $\nu_2(I_r)$.

The first step is a closed form for the binomial sum.

Lemma 3.2. *For $m \ge 0$ let*

$$J_m = I_{m+1} = \sum_{j=0}^{m} \binom{m}{j} \frac{1}{2j+1}, \qquad c_m = \frac{m!}{(2m+1)!!} = \frac{2^m}{(2m+1)\binom{2m}{m}}.$$

Then $(2m+1)J_m = 2^m + 2mJ_{m-1}$ for $m \ge 1$, and

$$J_m = c_m\, Q_m, \qquad Q_m = \sum_{k=0}^{m} 2^{m-k} \binom{2k}{k} \in \mathbb{Z}.$$

Proof. Since $\frac{d}{du}\big[u(1+u^2)^m\big] = (2m+1)(1+u^2)^m - 2m(1+u^2)^{m-1}$, integrating over $[0,1]$ gives $2^m = (2m+1)J_m - 2mJ_{m-1}$. The numbers c_m satisfy $(2m+1)c_m = mc_{m-1}$. Dividing the first relation by $(2m+1)c_m = mc_{m-1}$, the ratio $Q_m = J_m/c_m$ satisfies $Q_m = 2Q_{m-1} + 2^m/\big((2m+1)c_m\big) = 2Q_{m-1} + \binom{2m}{m}$, with $Q_0 = 1$; hence $Q_m = \sum_{k \le m} 2^{m-k}\binom{2k}{k}$. □

Next we show that Theorem 1.4 is equivalent to Theorem 1.3. We need the following elementary facts about the numbers c_k.

Lemma 3.3. *$\nu_2(c_k) = k - s_2(k) = \nu_2(k!)$. This is non-decreasing in k, because $\nu_2((k+1)!) = \nu_2(k!) + \nu_2(k+1)$, and tends to infinity. Hence each c_k belongs to $\mathbb{Z}_2$, every tail*

$$R_j = \sum_{k \ge j} c_k \qquad (j \ge 0)$$

converges in $\mathbb{Q}_2$, and $\nu_2(R_j) \ge j - s_2(j) = \nu_2(j!)$.

Proof. $2k+1$ is odd and $\nu_2\binom{2k}{k} = s_2(k)$ by Kummer's theorem, so $\nu_2(c_k) = k - s_2(k)$, which equals $\nu_2(k!)$ by Legendre's formula. □

Theorem 3.4. *The following are equivalent:*
(a) $\nu_2(J_m) = m + \nu_2(m+1)$ *for every* $m \geq 0$*;*
(b) $R_0 = \sum_{k\geq 0} c_k = 0$ *in* $\mathbb{Q}_2$.

Proof. A second expression for J_m. Since $(\frac{1}{2})_j/(\frac{3}{2})_j = 1/(2j+1)$, the sum J_m is the terminating hypergeometric series ${}_2F_1(-m, \frac{1}{2}; \frac{3}{2}; -1)$. Pfaff's transformation ${}_2F_1(a,b;c;z) = (1-z)^{-a}\,{}_2F_1\big(a, c-b; c; \frac{z}{z-1}\big)$ [13, Ch. 2] at $z=-1$ gives $J_m = 2^m\,{}_2F_1(-m, 1; \frac{3}{2}; \frac{1}{2})$. As $(-m)_k(1)_k 2^{-k}/\big((\frac{3}{2})_k k!\big) = (-1)^k\binom{m}{k} c_k$, we obtain

$$\widetilde{J}_m := \frac{J_m}{2^m} = \sum_{k=0}^{m} (-1)^k \binom{m}{k} c_k, \tag{15}$$

and (a) is the statement $\nu_2(\widetilde{J}_m) = \nu_2(m+1)$ for all m.

Summation by parts. Write $n = m+1$. Using $\binom{n-1}{k} = \sum_{j=0}^{k}(-1)^{k-j}\binom{n}{j}$ in (15) and exchanging the two summations (legitimate in $\mathbb{Q}_2$ because $c_k \to 0$),

$$\widetilde{J}_m = \sum_{j=0}^{n} (-1)^j \binom{n}{j} R_j = R_0 + n \sum_{j=1}^{n} \frac{(-1)^j}{j} \binom{n-1}{j-1} R_j, \tag{16}$$

where we used $\binom{n}{j} = \frac{n}{j}\binom{n-1}{j-1}$.

(b) implies (a). Assume $R_0 = 0$. Then $R_1 = -c_0 = -1$ and $R_2 = -c_0 - c_1 = -\frac{4}{3}$. In the last sum of (16), the term $j=1$ equals 1; the term $j=2$ is a multiple of $R_2/2 = -\frac{2}{3} \in 2\mathbb{Z}_2$; and for $j \geq 3$, Lemma 3.3 gives $\nu_2(R_j/j) \geq \nu_2(j!) - \nu_2(j) = \nu_2((j-1)!) \geq 1$. So that sum is 1 plus an element of $2\mathbb{Z}_2$, a unit, and $\nu_2(\widetilde{J}_m) = \nu_2(n) = \nu_2(m+1)$.

(a) implies (b). By (15) and $c_k \to 0$, the function $m \mapsto \widetilde{J}_m$ is given by a Mahler series, and therefore extends to a continuous function on $\mathbb{Z}_2$ [3, Ch. 4], whose value at $m=-1$ is $\sum_{k\geq 0}(-1)^k\binom{-1}{k} c_k = \sum_{k\geq 0} c_k = R_0$. The integers $2^k - 1$ tend to -1 in $\mathbb{Z}_2$ as $k \to \infty$, and by (a) $\nu_2(\widetilde{J}_{2^k-1}) = k \to \infty$. By continuity, $R_0 = 0$. □

It remains to prove the vanishing itself. For $i \geq 0$ consider the binomial moments of the sequence (c_k),

$$M_i = \sum_{k\geq i} c_k \binom{k}{i}.$$

By Lemma 3.3 these series converge in $\mathbb{Q}_2$, with

$$\nu_2(M_i) \geq i - s_2(i), \tag{17}$$

and $M_0 = R_0$ is the sum we want to evaluate.

Theorem 3.5. $M_0 = \displaystyle\sum_{k\geq 0} \frac{k!}{(2k+1)!!} = 0$ *in* $\mathbb{Q}_2$.

Proof. Step 1: a two-term recurrence for the moments. The relation $(2k+1)c_k = k c_{k-1}$ $(k \geq 1)$ gives, after shifting the index in the last sum,

$$\sum_k c_k (2k+1)\binom{k}{i} = \sum_k k c_{k-1}\binom{k}{i} = \sum_k (k+1) c_k \binom{k+1}{i},$$

that is, $\sum_k c_k\big[(2k+1)\binom{k}{i} - (k+1)\binom{k+1}{i}\big] = 0$. The elementary identities

$$\binom{k+1}{i} = \binom{k}{i} + \binom{k}{i-1}, \quad (k-i)\binom{k}{i} = (i+1)\binom{k}{i+1}, \quad (k-i+1)\binom{k}{i-1} = i\binom{k}{i}$$

show that the bracket equals $(i+1)\binom{k}{i+1} - i\binom{k}{i-1}$. Therefore

$$(i+1)M_{i+1} = iM_{i-1} \qquad (i \geq 1). \tag{18}$$

Step 2: the even moments are explicit multiples of M_0. Taking $i = 2j-1$ in (18) gives $2jM_{2j} = (2j-1)M_{2j-2}$, hence

$$M_{2j} = M_0 \prod_{t=1}^{j} \frac{2t-1}{2t} = \frac{M_0}{4^j}\binom{2j}{j}, \qquad \text{so} \qquad \nu_2(M_{2j}) = \nu_2(M_0) + s_2(j) - 2j.$$

Step 3: comparison with the a priori bound. By (17) with $i = 2j$, $\nu_2(M_{2j}) \geq 2j - s_2(2j) = 2j - s_2(j)$. Combining with Step 2,

$$\nu_2(M_0) \;\geq\; 4j - 2s_2(j) \qquad \text{for every } j \geq 1.$$

The right-hand side is unbounded as $j \to \infty$, so $\nu_2(M_0) = +\infty$, that is, $M_0 = 0$. □

In words: if $M_0 \neq 0$, the recurrence would force $\nu_2(M_{2j})$ to tend to $-\infty$, whereas the direct convergence estimate forces it to tend to $+\infty$.

Proof of Theorems 1.3 and 1.4. Theorem 1.3 is Theorem 3.5, and Theorem 1.4 then follows from Theorem 3.4. □

Remark 3.6. The partial sums illustrate the vanishing: $\nu_2\big(\sum_{k<K} c_k\big) = 9$, 20, 48, 99, 200, 401, 802 for $K = 10$, 20, 50, 100, 200, 400, 800. Since the full sum is 0, the partial sum equals $-R_K$, and these values are just above the lower bound $K - s_2(K)$ of Lemma 3.3. For related work on p-adic valuations of binomial sums see [11, 12].

Finally, we apply these results to the phase constants.

Corollary 3.7. *For every r for which (14) holds, in particular for $r \leq 20$,*

$$\nu_2\big([\mu_\alpha^r]\kappa_r\big) = -2r - \nu_2\big((r-1)!\big) = -E_r + (r-1).$$

Proof. By Theorem 1.4 with $m = r-1$, $\nu_2(I_r) = (r-1) + \nu_2(r)$. By Kummer's theorem $\nu_2(\gamma_r) = 1 + s_2(r) - 4r$. Adding, and using $\nu_2(r) = 1 - s_2(r) + s_2(r-1)$ and $\nu_2((r-1)!) = r - 1 - s_2(r-1)$, we get $-3r + 1 + s_2(r-1) = -2r - \nu_2((r-1)!)$. □

Remark 3.8 (a gap of $r-1$ in the shifted variables). Let δ_r be the exponent of 2 in the common denominator of the coefficients of κ_r written in the variables (μ_α, μ_β). Theorem 1.2 and Corollary 3.7 give, whenever (14) holds,

$$2r + \nu_2\big((r-1)!\big) \;\leq\; \delta_r \;\leq\; 3r - 1 + \nu_2\big((r-1)!\big).$$

The two bounds differ by $r-1$. Determining δ_r exactly is an open problem.

4. The sharp law as a property of one power series

We now turn to the coefficients $\ell_r = [A^1B^0]\kappa_r$ of (3) and prove Theorem 1.5. Let us first explain where the difficulty lies. By (13), ℓ_r is a sum of a large number of terms, indexed by the monomials $x^c y^d \mu_\alpha^a \mu_\beta^b$ of T_{2r-1}. The sharp law (4) cannot be read off from these terms: the valuation of the sum is strictly larger than the minimum of the valuations of the terms, and the same happens for every natural way we tried of grouping the terms into partial sums (for instance by the degree $c+d$). The 2-adic cancellation is a global property of the sum. The construction below first performs the whole summation, in the form of a generating function, and only then extracts valuations.

We now introduce the generating function on which the argument is based. Let L_n be as in Section 2, and define

$$g_n = \frac{\partial L_n}{\partial A}(x_0, y_0)\Big|_{A=B=0} \in \mathbb{Q}(i) \qquad (n \geq 1). \tag{19}$$

By Definition 2.1 and (3),

$$\ell_r = 2^{-(2r-1)} \operatorname{Im} g_{2r-1}. \tag{20}$$

For example $L_1 = x\mu_\alpha + y\mu_\beta$ gives $g_1 = x_0 = \frac{1+i}{2}$ and $\ell_1 = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}$. The numbers g_n grow like $(n-1)!$, so we use the exponential-type generating function (Borel transform)

$$\Phi(t) = \sum_{n\geq 1} \frac{g_n}{(n-1)!}\, t^n, \qquad \operatorname{Im}\Phi(t) = \sum_{n\geq 1} \frac{\operatorname{Im} g_n}{(n-1)!}\, t^n,$$

where $\operatorname{Im}\Phi$ denotes the series of the imaginary parts of the coefficients.

Definition 4.1.

$$W(t) = \frac{\sinh 2t}{t^2}\, \operatorname{Im}\Phi(t), \qquad w_m = [t^{2m}]\, W(t) \quad (m \geq 0). \tag{21}$$

The first coefficients are

$$\sum_{m\geq 0} w_m t^{2m} = 1 + \frac{t^2}{12} + \frac{t^4}{64} - \frac{143\, t^6}{80640} + \frac{21787\, t^8}{46448640} - \cdots$$

Only the coefficients of even powers of t in W are used. (In fact W is even; see Remark 4.5.)

The proof uses two auxiliary series, whose coefficients have simple valuations.

Lemma 4.2. *Let*

$$S(t) = \frac{\sinh 2t}{2t} = \sum_{k\geq 0} e_k t^{2k}, \qquad D(t) = \frac{2t}{\sinh 2t} = \sum_{k\geq 0} d_k t^{2k},$$

so that $D(t)S(t) = 1$. *Then* $d_0 = e_0 = 1$ *and* $\nu_2(d_k) = \nu_2(e_k) = s_2(k)$ *for* $k \geq 1$.

Proof. We have $e_k = 2^{2k}/(2k+1)!$, and Legendre's formula with $s_2(2k+1) = s_2(k) + 1$ gives $\nu_2(e_k) = 2k - \big(2k + 1 - s_2(k) - 1\big) = s_2(k)$. For D, the classical expansion $u/\sinh u = \sum_k (2 - 2^{2k})\mathcal{B}_{2k} u^{2k}/(2k)!$ (obtained from [2, Eq. 4.19.4] by $z \mapsto iu$) at $u = 2t$ gives $d_k = (2 - 2^{2k})\mathcal{B}_{2k} 2^{2k}/(2k)!$. For $k \geq 1$ we have $\nu_2(2 - 2^{2k}) = 1$, $\nu_2(\mathcal{B}_{2k}) = -1$ by the von Staudt–Clausen theorem, and $\nu_2((2k)!) = 2k - s_2(k)$; hence $\nu_2(d_k) = 1 - 1 + 2k - (2k - s_2(k)) = s_2(k)$. □

We are now ready to prove Theorem 1.5. Put $\tau(m) = -2\nu_2\big((2m)!\big) = -4m + 2s_2(m)$.

Lemma 4.3 (convolutions that preserve τ)**.** *Let $(h_k)_{k\ge 0}$ be rational numbers with $\nu_2(h_0) = 0$ and $\nu_2(h_k) \ge s_2(k)$ for $k \ge 1$; let $(a_m)_{m\ge 0}$ be rational numbers and $a'_m = \sum_{k=0}^{m} h_k a_{m-k}$. Fix $M \ge 0$ and assume $\nu_2(a_j) \ge \tau(j)$ for all $j < M$. Then $\nu_2(a'_j) \ge \tau(j)$ for all $j < M$, and*

$$\nu_2(a'_M) \ge \tau(M) \iff \nu_2(a_M) \ge \tau(M), \qquad \nu_2(a'_M) = \tau(M) \iff \nu_2(a_M) = \tau(M).$$

Proof. For $1 \le k \le m$, the subadditivity $s_2(m) \le s_2(m-k) + s_2(k)$ gives

$$\big[s_2(k) + \tau(m-k)\big] - \tau(m) = s_2(k) + 4k + 2s_2(m-k) - 2s_2(m) \;\ge\; 4k - s_2(k) \;>\; 0.$$

Hence, if $\nu_2(a_j) \ge \tau(j)$ for $j < m$, every term $h_k a_{m-k}$ with $k \ge 1$ has valuation strictly larger than $\tau(m)$, so $a'_m = h_0 a_m +$ (an element of valuation $> \tau(m)$) with h_0 a unit. All assertions follow from the rule for the valuation of a sum (recalled in the Introduction). □

Proof of Theorem 1.5. By Definition 4.1, $W(t) = S(t) \cdot 2\operatorname{Im}\Phi(t)/t$. Since S is even, the coefficients of the even powers on both sides give

$$w_m = \sum_{k=0}^{m} e_k\, b_{m-k}, \qquad b_m = [t^{2m}]\frac{2\operatorname{Im}\Phi(t)}{t} = \frac{2\operatorname{Im} g_{2m+1}}{(2m)!},$$

and conversely, since $DS = 1$, $b_m = \sum_{k=0}^{m} d_k w_{m-k}$. By (20), $b_m = 2^{2m+2}\ell_{m+1}/(2m)!$, so

$$\nu_2(b_m) = 2m + 2 + \nu_2(\ell_{m+1}) - \big(2m - s_2(m)\big) = 2 + s_2(m) + \nu_2(\ell_{m+1}).$$

Since $E_{m+1} = 4m + 2 - s_2(m)$ by Legendre's formula, this shows that

$$\nu_2(b_m) \ge \tau(m) \iff \nu_2(\ell_{m+1}) \ge -E_{m+1},$$

and likewise with equalities. It remains to pass between (b_m) and (w_m): by Lemma 4.2, both (e_k) and (d_k) satisfy the hypotheses of Lemma 4.3, which, applied with $h = e$, $a = b$ and with $h = d$, $a = w$, gives the equivalence in both directions, with $M = r - 1$. Finally, $\nu_2(w_m) = \tau(m) = -2\nu_2((2m)!)$ says exactly that $((2m)!)^2 w_m$ is a unit of $\mathbb{Z}_2$. □

Remark 4.4 (what has been gained). The sharp law was a statement about sums over the four-index family of coefficients of T_{2r-1}. After Theorem 1.5 it is a statement about the Taylor coefficients of one function of one variable, with a normalisation, $((2m)!)^2$, which does not involve the irregular quantity E_r. The passage from Φ to W mixes infinitely many coefficients, yet loses no 2-adic information, because in each convolution one term strictly dominates all the others. It is in this form that the law is proved in [17].

Remark 4.5 (the real parts). The real parts of the g_n are much simpler than the imaginary parts. In the computed range $n \le 95$ one finds, exactly,

$$\operatorname{Re}\Phi(t) = \frac{t^2 e^t}{\sinh 2t}, \qquad \operatorname{Im} g_{2k} = 0,$$

so that W is even and $\frac{\sinh 2t}{t^2}\,\Phi(t) = e^t + i\,W(t)$. Equivalently, $\operatorname{Re} g_{2k+1} = (1 - 2^{2k-1})\mathcal{B}_{2k}$ and $\operatorname{Re} g_{2k} = \frac{1}{2}(2k-1)\mathcal{E}_{2k-2}$, where $\mathcal{E}_{2k}$ are the Euler numbers, which are odd integers; thus $\nu_2(\operatorname{Re} g_n) = -1$ for every n. These identities are proved in [17]; they are not used in the present paper.

m	$\nu_2(w_m)$	$u_m = \big((2m)!\big)^2 w_m$
0	0	1
1	−2	1/3
2	−6	9
3	−8	−6435/7
4	−14	762545
5	−16	−18081757575/11
6	−20	8219311479225
7	−22	−82171171579421925
8	−30	1483464568148296358625
9	−32	−85169503555943977729796737 5/19
10	−36	213796566490722467733033615 5625

TABLE 2. The numbers u_m of property (W). Each is a quotient of two odd integers, hence a unit of $\mathbb{Z}_2$, and $\nu_2(w_m) = -2\nu_2((2m)!)$.

5. Computations

This section reports the exact computations that accompany the results above. We begin with the verification of the sharp law. The numbers g_n were computed exactly for $n \le 95$ from the recurrence of Theorem 2.3. Since only the derivative in A at $A = B = 0$ is needed, the recurrence is run with $\mu_\alpha = -\frac{1}{4} + \varepsilon$, $\mu_\beta = -\frac{1}{4}$, discarding ε^2; after multiplication by 4^n all quantities are integers.

Computational observation 5.1. $\nu_2(\ell_r) = -E_r$ for $r \le 48$, by direct computation of ℓ_r; and $\nu_2(w_m) = -2\nu_2\big((2m)!\big)$ for $m \le 46$, which corresponds to $r \le 47$ in Theorem 1.5.

Table 2 lists the first values of $u_m = ((2m)!)^2 w_m$ after removal of the power of 2; property (W) asserts that this power is 2^0, i.e. that $\nu_2(w_m) = -2\nu_2((2m)!)$, which is the second column.

Remark 5.2. An element of $\mathbb{Z}_2$ is a unit if and only if it is congruent to 1 modulo 2. With $\mathcal{U}(q) = \sum_{m\ge 0} u_m q^m$, property (W) therefore says that $\mathcal{U}$ has coefficients in $\mathbb{Z}_2$ and satisfies the single congruence $(1-q)\,\mathcal{U}(q) \equiv 1 \pmod 2$.

We finally discuss the absence of a simple equation for W. A standard way to prove the integrality of the coefficients of a power series is to find a linear differential equation or recurrence that it satisfies and to study it p-adically, in the manner of Dwork [9, 10]. Within the finite family of equations described below, we found none.

Computational observation 5.3. With w_m known exactly for $m \le 46$, none of the three series

$$\sum_m w_m q^m, \qquad \sum_m \big((2m)!\big)^2 w_m q^m, \qquad \sum_m \frac{\operatorname{Im} g_{2m+1}}{(2m)!} q^m$$

satisfies any of the following, with polynomial coefficients of degree at most D and $(R+1)(D+1) \le 35$: a linear recurrence $\sum_{i\le R} p_i(m) a_{m+i} = 0$; a linear differential equation $\sum_{i\le R} p_i(q) f^{(i)} = 0$; an algebraic equation $\sum_{i\le R} p_i(q) f^i = 0$. Each search was made with at least six more equations than unknowns. The same search recovers the known equations for $\sum q^m/((2m)!)^2$, for the generating function of the Catalan numbers and for $\sum m!\, q^m$.

This does not exclude that W satisfies an equation of larger size. It shows only that no relation occurs within the tested family; a relation of larger order or degree, or of a different type, is not ruled out. The proof in [17] does not rely on such an equation.

Appendix A. An integral representation

This appendix is independent of the rest of the paper. In [17] the numbers g_n are obtained from the Gauss hypergeometric function ${}_2F_1(a,b;a+b;z)$ at the point $z=2$, which lies on its branch cut $[1,\infty)$; the value there is understood as the limit from the upper half-plane, written ${}_2F_1^{\uparrow}(a,b;a{+}b;2)$. More precisely, with $\sigma=(1-Z)/2$, $a=\sigma-\delta/2$ and $b=\sigma+\delta/2$, the generating function of the g_n is obtained from the second derivative of this boundary value with respect to δ at $\delta=0$. We record an elementary formula which separates the contribution of the branch cut. Below, $\mathrm{B}(p,q)=\Gamma(p)\Gamma(q)/\Gamma(p+q)$ is the Beta function.

Proposition A.1. *Assume* $0<\operatorname{Re}a<1$ *and* $\operatorname{Re}b>0$. *Then*

$$
{}_2F_1^{\uparrow}(a,b;a+b;2)=\frac{1}{\mathrm{B}(a,b)}\Big\{2^{-b}\int_0^1 u^{b-1}(1-u/2)^{a-1}(1-u)^{-a}\,du \\
+e^{i\pi a}2^{1-a-b}\int_0^1 u^{-a}(1-u)^{a-1}(1+u)^{b-1}\,du\Big\},
$$

and

$$
\int_0^1 u^{b-1}(1-u/2)^{a-1}(1-u)^{-a}\,du=\mathrm{B}(b,1-a)\,{}_2F_1\big(1-a,b;\,b+1-a;\,\tfrac12\big),
$$

$$
\int_0^1 u^{-a}(1-u)^{a-1}(1+u)^{b-1}\,du=\mathrm{B}(1-a,a)\,{}_2F_1\big(1-b,1-a;\,1;\,-1\big).
$$

Both sides extend meromorphically in (a,b), *and the identity persists. At* $\delta=0$, *that is* $a=b=\sigma$, *the two Gauss functions are*

$$
{}_2F_1\big(1-\sigma,\sigma;1;\tfrac12\big)=\frac{\sqrt{\pi}}{\Gamma\big(1-\frac{\sigma}{2}\big)\Gamma\big(\frac{\sigma+1}{2}\big)},\qquad {}_2F_1\big(1-\sigma,1-\sigma;1;-1\big)=\frac{\Gamma\big(\frac{3-\sigma}{2}\big)}{\Gamma(2-\sigma)\,\Gamma\big(\frac{1+\sigma}{2}\big)}.
$$

Proof. Euler's integral ${}_2F_1(a,b;a+b;z)=\mathrm{B}(a,b)^{-1}\int_0^1 v^{b-1}(1-v)^{a-1}(1-zv)^{-a}dv$ is valid for z in the upper half-plane; let $z\to 2$ (dominated convergence applies under the stated assumptions) and split the integral at $v=\frac12$. On $(0,\frac12)$ put $v=u/2$. On $(\frac12,1)$ put $v=(1+u)/2$; there $1-zv$ tends to the negative number $-u$ from below, so $(1-zv)^{-a}\to e^{i\pi a}u^{-a}$. This gives the first formula. The two integrals are again Euler integrals, $\int_0^1 u^{q-1}(1-u)^{c-q-1}(1-zu)^{-p}du=\mathrm{B}(q,c-q)\,{}_2F_1(p,q;c;z)$, with $(p,q,c,z)=(1-a,b,b+1-a,\frac12)$ and $(1-b,1-a,1,-1)$ respectively. At $a=b=\sigma$ the first is evaluated by Gauss's second summation theorem and the second by Kummer's summation theorem [13, Sec. 3.1]. □

Thus the boundary value is the sum of a part without branch contribution and a part carrying the explicit phase $e^{i\pi a}$, and at $\delta=0$ both parts are quotients of Gamma functions. The proposition also shows that replacing the upper boundary value by the lower one conjugates the phase, so the choice matters. For background on the functions ${}_2F_1(a,b;a+b;z)$ see [15, 16]; the reduction of the double series (9) to a Gauss function is of the type classified in [14].

Declarations

Competing interests. The author declares no competing interests.

Data availability. This paper reports no experimental data. All computations are exact, in rational or Gaussian-rational arithmetic.

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